

CHEMISTRY**PAPER 1 (THEORY)****Maximum Marks: 70****Time Allowed: Three Hours***Reading Time: Additional Fifteen Minutes***Instructions to Candidates**

1. You are allowed an additional fifteen minutes for only reading the question paper.
2. You must NOT start writing during the reading time.
3. This question paper has 9 printed pages.
4. It is divided into four sections and has twenty one questions in all.
5. Answer all questions.
6. Section A has fourteen subparts. Each question carries 1 mark.
7. While attempting Multiple Choice Questions in Section A, you are required to write only ONE option as the answer.
8. Section B has ten questions. Each question carries 2 marks.
9. Section C has seven questions. Each question carries 3 marks.
10. Section D has three questions. Each question carries 5 marks.
11. Internal choices have been provided in two questions each in Sections B, C and D.
12. The intended marks for questions are given in brackets [].
13. All working, including rough work, should be done on the same sheet as, and adjacent to the rest of the answer.
14. Balanced equations must be given wherever possible and diagrams where they are helpful.
15. When solving numerical problems, all essential workings must be shown.
16. In working out problems, use the following data:
Gas constant $R = 1.987 \text{ cal deg}^{-1} \text{ mol}^{-1} = 8.314 \text{ J K}^{-1} \text{ mol}^{-1} = 0.0821 \text{ dm}^3 \text{ atm K}^{-1} \text{ mol}^{-1}$, $1 \text{ l atm} = 1 \text{ dm}^3 \text{ atm} = 101.3 \text{ J}$.
 $1 \text{ faraday} = 96500 \text{ coulombs}$, Avogadro's number $= 6.023 \times 10^{23}$.

Instruction to Supervising Examiner*Kindly read aloud the Instructions given above to all the candidates present in the examination hall.***SECTION A – 14 MARKS****Question 1**

- (A) Fill in the blanks by choosing the appropriate word(s) from those given in the brackets.
[3-hydroxybutanal, 2-hydroxybutanal, aldol, Cannizzaro, 96500, 193000, two, three, nearly identical, decreases, increases, salicylaldehyde, salicylic acid, Reimer-Tiemann, Kolbe's, anode]
- (i) Two molecules of acetaldehyde condense in the presence of dilute alkali to give _____, [1]
which loses water on heating to form but-2-enal. The reaction is known as _____
condensation. (Recall)

- (ii) The quantity of electricity needed to reduce one mole of Cu^{2+} ions to copper is _____ coulombs, whereas one mole of Al^{3+} ions needs _____ faradays to be reduced to aluminium. (*Apply*) [1]
- (iii) As a result of lanthanoid contraction, the atomic radii of zirconium and hafnium are _____, and the basic strength of the hydroxides _____ from $\text{La}(\text{OH})_3$ to $\text{Lu}(\text{OH})_3$. (*Understand*) [1]
- (iv) Phenol, on treatment with chloroform and aqueous sodium hydroxide followed by acidification, gives _____. The reaction is known as _____ reaction. (*Recall*) [1]
- (B) Select and write the correct alternative from the choices given below.
- (i) Consider 0.1 m aqueous solutions of glucose, NaCl and K_2SO_4 at the same temperature (assume complete dissociation of the salts). [1]
- P: The K_2SO_4 solution has the lowest freezing point.
 Q: The glucose and NaCl solutions have the same osmotic pressure.
 R: The van't Hoff factor of NaCl is 2.
 S: The elevation of boiling point of the K_2SO_4 solution is three times that of the glucose solution.
- Which of the statements are correct?
 (*Evaluate*)
- (a) P, Q and R only
 (b) P, R and S only
 (c) Q, R and S only
 (d) P, Q and S only
- (ii) On hydrolysis, lactose gives: (*Recall*) [1]
- (a) α -D-glucose and β -D-fructose
 (b) two molecules of α -D-glucose
 (c) β -D-galactose and β -D-glucose
 (d) β -D-galactose and α -D-fructose
- (iii) One mole of $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$, on treatment with excess of silver nitrate solution, precipitates two moles of AgCl. The correct formula of the complex is: (*Understand*) [1]
- (a) $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$
 (b) $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$
 (c) $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$
 (d) $[\text{Cr}(\text{H}_2\text{O})_3\text{Cl}_3] \cdot 3\text{H}_2\text{O}$

- (iv) For the reaction $A + B \rightarrow \text{products}$ the following data were obtained: [1]

Run	[A] (mol L ⁻¹)	[B] (mol L ⁻¹)	Initial rate (mol L ⁻¹ s ⁻¹)
1	0.10	0.10	2.0×10^{-3}
2	0.20	0.10	8.0×10^{-3}
3	0.10	0.20	4.0×10^{-3}

The rate law for the reaction is:

(Apply)

- (a) rate = $k[A][B]$
(b) rate = $k[A][B]^2$
(c) rate = $k[A]^2[B]$
(d) rate = $k[A]^2[B]^0$
- (v) The correct order of basic strength of the following in aqueous solution is: (Analyse) [1]
- (a) $(\text{CH}_3)_3\text{N} > (\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2 > \text{NH}_3$
(b) $(\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2 > (\text{CH}_3)_3\text{N} > \text{NH}_3$
(c) $\text{CH}_3\text{NH}_2 > (\text{CH}_3)_2\text{NH} > (\text{CH}_3)_3\text{N} > \text{NH}_3$
(d) $\text{NH}_3 > \text{CH}_3\text{NH}_2 > (\text{CH}_3)_2\text{NH} > (\text{CH}_3)_3\text{N}$
- (vi) Given below are two statements marked Assertion and Reason. Read the two statements carefully and select the correct option. [1]

Assertion: Di-*tert*-butyl ether cannot be prepared by Williamson's synthesis from sodium *tert*-butoxide and *tert*-butyl bromide.

Reason: With a strongly basic alkoxide, a tertiary alkyl halide undergoes elimination to give an alkene rather than substitution.

(Analyse)

- (a) Both Assertion and Reason are true, and Reason is the correct explanation for Assertion.
(b) Both Assertion and Reason are true, but Reason is not the correct explanation for Assertion.
(c) Assertion is true and Reason is false.
(d) Both Assertion and Reason are false.
- (vii) Given below are two statements marked Assertion and Reason. Read the two statements carefully and select the correct option. [1]

Assertion: The molar conductivity of a weak electrolyte increases steeply on dilution.

Reason: The number of ions per unit volume of the solution increases on dilution.

(Analyse)

- (a) Both Assertion and Reason are true, and Reason is the correct explanation for Assertion.
(b) Both Assertion and Reason are true, but Reason is not the correct explanation for Assertion.
(c) Assertion is true and Reason is false.
(d) Both Assertion and Reason are false.

- (C) Read the passage carefully and answer the questions that follow.

Radioactive decay is a first order process: the number of nuclei that decay per unit time is proportional to the number present, $N = N_0 e^{-\lambda t}$, and the half-life $t_{1/2} = 0.693/\lambda$. Iodine-131 ($t_{1/2} = 8$ days) is used to treat an overactive thyroid gland, and carbon-14 ($t_{1/2} = 5730$ years) to date archaeological samples.

A living plant keeps the ratio of ^{14}C to ^{12}C in its tissues constant by photosynthesis. Once it dies, no fresh carbon-14 is taken in, and the activity of the ^{14}C in the sample falls to one half every 5730 years.

- (i) Why is the half-life of a radioactive isotope independent of the amount of the isotope present? (Understand) [1]
- (ii) What fraction of a dose of iodine-131 remains after 24 days? (Apply) [1]
- (iii) A piece of old wood shows a ^{14}C activity that is 25 % of that of fresh wood. Estimate its age. (Apply) [1]

SECTION B – 20 MARKS

Question 2

The standard reduction potentials of three electrodes are given below.

[2]

Electrode	E° (V)
Zn^{2+}/Zn	-0.76
Cu^{2+}/Cu	+0.34
Ag^+/Ag	+0.80

A student wishes to store (a) 1 M silver nitrate solution and (b) 1 M zinc sulphate solution, each in a copper vessel. Predict, with reason, whether each can be stored safely.

(Apply)

Question 3

An alkene [A] with molecular formula C_6H_{12} on ozonolysis followed by hydrolysis in the presence of zinc gives a single compound [B]. Compound [B] gives a yellow precipitate with iodine and sodium hydroxide but does not reduce Fehling's solution. [B] on reduction with zinc amalgam and concentrated hydrochloric acid gives a gas [C]. Identify [A], [B] and [C] and write the equation for the formation of [C] from [B]. (Understand)

[2]

Question 4

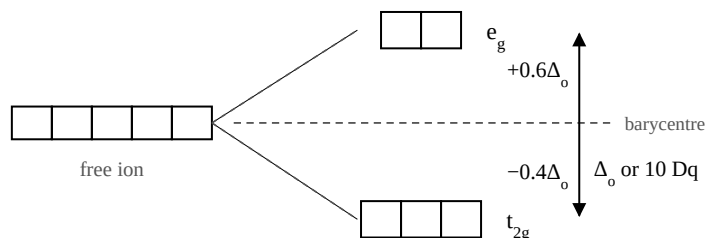
- (i) Answer the following questions with respect to the complex ion $[\text{Fe}(\text{CN})_6]^{3-}$ (atomic number of Fe = 26).
- (a) What is the hybridisation state of the central metal ion? (Understand) [1]

(b) Comment on the magnetic nature of the complex ion. (*Analyse*)

[1]

OR

- (ii) Observe the diagram below of splitting of d orbitals in an octahedral field and answer the questions that follow.



- (a) Write the electronic configuration of a d^6 ion if $\Delta_o > P$, and state the number of unpaired electrons. (*Understand*)

[1]

- (b) Calculate the crystal field stabilisation energy (CFSE) of this configuration in terms of Δ_o . (*Apply*)

[1]

Question 5

The decomposition of N_2O_5 is a first order reaction. If 30 % of it decomposes in 15 minutes, calculate the time required for 75 % of it to decompose. (*Apply*)

[2]

Question 6

How can the following conversions be carried out? (Write the chemical equations.)

- (i) Aniline to phenol (*Recall*)

[1]

- (ii) Chlorobenzene to biphenyl (*Recall*)

[1]

Question 7

Substantiate the following statements with reasons.

- (i) Sc^{3+} salts are colourless whereas Ti^{3+} salts are coloured. (Atomic number of Sc = 21, Ti = 22) (*Understand*)

[1]

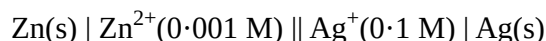
- (ii) Zinc is not regarded as a transition element. (Atomic number of Zn = 30) (*Understand*)

[1]

Question 8

Write the Nernst equation and calculate the emf of the following cell at 298 K:

[2]



Given: $E^\circ(Zn^{2+}/Zn) = -0.76 V$, $E^\circ(Ag^+/Ag) = +0.80 V$ (*Apply*)

Question 9

- (i) Illustrate the following reactions with an example in each case:

- (a) Hofmann bromamide degradation (*Recall*)

[1]

(b) Gattermann reaction (*Recall*) [1]

OR

- (ii) An aromatic compound [A], C_7H_7NO , on heating with bromine and potassium hydroxide gives [B], C_6H_7N . [B] reacts with sodium nitrite and hydrochloric acid at 273–278 K to give [C], which couples with phenol in alkaline medium to give an orange dye [D]. Write the structures of [A], [B], [C] and [D]. (*Understand*) [2]

Question 10

The rate constant of a reaction is $2.0 \times 10^{-2} \text{ s}^{-1}$ at 300 K and $8.0 \times 10^{-2} \text{ s}^{-1}$ at 320 K. Calculate the energy of activation of the reaction. (*Apply*) [2]

Question 11

In a hill town the winter temperature falls to -5°C . Calculate the mass of ethylene glycol ($C_2H_6O_2$) that must be added to 4.0 kg of water in a car radiator so that the water does not freeze at this temperature. (Given: K_f of water = $1.86 \text{ K kg mol}^{-1}$; C = 12, H = 1, O = 16) (*Apply*) [2]

SECTION C – 21 MARKS

Question 12

Two isomeric bromides, [X] and [Y], of molecular formula C_4H_9Br were hydrolysed with aqueous KOH. The rate of hydrolysis of [X] was found to depend only on the concentration of [X], while that of [Y] depended on the concentrations of both [Y] and OH^- . [Y] is a straight-chain primary bromide.

Meera concluded that [X] is 1-bromobutane because it reacts faster; Arjun concluded that [X] is 2-bromo-2-methylpropane.

- (i) Whose conclusion is correct? Write the structures and IUPAC names of [X] and [Y]. (*Evaluate*) [1]
- (ii) Write the rate law and name the mechanism followed by each. (*Understand*) [1]
- (iii) Why does [X] follow the mechanism you have named, and not the other one? (*Analyse*) [1]

Question 13

Write the chemical equation for each of the following reactions:

- (i) Rosenmund reduction (*Recall*) [1]
- (ii) Wolff–Kishner reduction (*Recall*) [1]
- (iii) Hell–Volhard–Zelinsky reaction (*Recall*) [1]

Question 14

- (i) Two isomeric alcohols [P] and [Q] have the molecular formula C_3H_8O .
- (a) Draw their structures and write their IUPAC names. (*Understand*) [1]
- (b) Which of them gives the iodoform test? Write the balanced equation. (*Apply*) [1]

- (ii) Ethanol ($M = 46$) boils at 351 K whereas methoxymethane ($M = 46$) boils at 248 K. Explain. [1]
(Understand)

OR

- (iii) How will the following be converted? (Give chemical equations.)
- (a) Phenol to anisole (Recall) [1]
- (b) 2-Methylpropan-2-ol from a Grignard reagent (Recall) [1]
- (c) Ethoxyethane from ethanol (Recall) [1]

Question 15

- (i) Write the IUPAC names of: (a) $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}_2$ (b) $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$ (Apply) [1]
- (ii) Write the formula of the linkage isomer of the complex in (a) and give a reason why such isomerism is possible. (Understand) [1]
- (iii) What type of isomerism is shown by $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ and $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$? How can they be distinguished by a chemical test? (Analyse) [1]

Question 16

- (i) Glucose is oxidised by bromine water. Name the product and state what this reaction shows about the structure of glucose. (Analyse) [1]
- (ii) What is a peptide bond? Draw the structure of the dipeptide glycylalanine and mark the peptide bond. (Understand) [1]
- (iii) Vitamin C must be supplied regularly in the diet, whereas vitamin A need not be. Explain. (Understand) [1]

Question 17

- (i) Two electrolytic cells containing solutions of CuSO_4 and AgNO_3 respectively are connected in series. A steady current of 0.5 ampere is passed until 2.16 g of silver is deposited at the cathode of the second cell. (Atomic weight of Ag = 108, Cu = 63.5)
- (a) What mass of copper is deposited? (Apply) [1½]
- (b) For how long was the current passed? (Apply) [1½]

OR

- (ii) The molar conductivity of $0.001028 \text{ mol L}^{-1}$ acetic acid is $48.15 \text{ S cm}^2 \text{ mol}^{-1}$ at 298 K. The limiting ionic conductivities of H^+ and CH_3COO^- are 349.6 and $40.9 \text{ S cm}^2 \text{ mol}^{-1}$ respectively.
- (a) State Kohlrausch's law of independent migration of ions. (Recall) [1]
- (b) Calculate the degree of dissociation and the dissociation constant of acetic acid at this concentration. (Apply) [2]

Question 18

Write the chemical equations for the following conversions: (Recall)

- (i) Acetamide to methanamine [1]
- (ii) Aniline to fluorobenzene [1]
- (iii) Ethyl bromide to propan-1-amine [1]

SECTION D – 15 MARKS

Question 19

- (i) Read the passage and answer the questions that follow.

Phenol, C_6H_5OH , is manufactured mainly from cumene, which is oxidised by air to cumene hydroperoxide; the hydroperoxide is then cleaved by dilute acid. Phenol is a weak acid, stronger than ethanol and water: it turns blue litmus red and dissolves in aqueous NaOH, but not in $NaHCO_3$. The $-OH$ group activates the ring so strongly that phenol decolourises bromine water at once, giving a white precipitate. A dilute (0.2 %) solution of phenol is an antiseptic and a 1 % solution is a disinfectant.

- (a) Write the chemical equations for the manufacture of phenol from cumene. (*Recall*) [1]
(b) Why is phenol a stronger acid than ethanol? (*Evaluate*) [1]
(c) Write the equation for the reaction of phenol with bromine water and name the product. (*Apply*) [1]
(d) Arrange in increasing order of acid strength: ethanol, 4-nitrophenol, phenol, 4-methylphenol. (*Analyse*) [1]
(e) Give a chemical test to distinguish between phenol and ethanol. (*Apply*) [1]

OR

- (ii) Answer the following:

- (a) Chlorobenzene is heated with aqueous NaOH at 623 K and 300 atm to give [A], which on acidification gives [B]. [B] with chloroform and aqueous NaOH, followed by acidification, gives [C]. The sodium salt of [B] heated with CO_2 at 400 K and 4–7 atm, followed by acidification, gives [D]. Identify [A], [B], [C] and [D]. (*Analyse*) [2]
(b) Anisole on heating with HI gives phenol and methyl iodide, not iodobenzene and methanol. Explain. (*Analyse*) [1]
(c) 2-Nitrophenol is steam volatile but 4-nitrophenol is not. Why? (*Evaluate*) [1]
(d) Why should ethers not be distilled to dryness? (*Understand*) [1]

Question 20

- (i) Complete and balance the following chemical equations:

- (a) $K_2Cr_2O_7 + H_2SO_4 + FeSO_4 \rightarrow \text{____} + \text{____} + \text{____} + \text{____}$ (*Understand*) [1]
(b) $KMnO_4 + H_2SO_4 + H_2C_2O_4 \rightarrow \text{____} + \text{____} + \text{____} + \text{____}$ (*Understand*) [1]

- (ii) Give reasons for the following:

- (a) Transition metals and their compounds are good catalysts. (*Understand*) [1]
(b) An orange solution of potassium dichromate turns yellow when sodium hydroxide is added to it. (*Analyse*) [1]
(c) Among the bivalent ions of the first transition series, Mn^{2+} shows the maximum paramagnetic character. (Atomic number of Mn = 25) (*Analyse*) [1]

Question 21

(i) Answer the following:

- (a) A solution of 12.48 g of barium chloride in 1.0 kg of water boils at 373.0832 K. Calculate the degree of dissociation of barium chloride. (K_b of water = $0.52 \text{ K kg mol}^{-1}$, boiling point of water = 373 K, molar mass of $\text{BaCl}_2 = 208 \text{ g mol}^{-1}$) (*Apply*) [2]
- (b) The vapour pressure of water at 293 K is 17.535 mm Hg. Calculate the vapour pressure of water at 293 K when 25 g of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) is dissolved in 450 g of water. (*Apply*) [2]
- (c) Why are bottles of soda water sealed under high pressure? (*Understand*) [1]

OR

(ii) Answer the following:

- (a) A mixture of ethanol and water shows positive deviation from Raoult's law. Explain the reason, name the type of azeotrope it forms and sketch the vapour pressure–composition curve for such a solution. (*Evaluate*) [2]
- (b) Calculate the molality and the mole fraction of the solute in a 20 % (by mass) aqueous solution of ethylene glycol ($\text{C}_2\text{H}_6\text{O}_2$). ($\text{C} = 12, \text{H} = 1, \text{O} = 16$) (*Apply*) [2]
- (c) Why is molality preferred to molarity in the study of colligative properties? (*Understand*) [1]

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